CCIT4080 Project on Knowledge Products Development - Rubrics

Learning outcome	Excellent	Good	Satisfactory	Fair	Poor
Identify practical, real-life problem with accessible learning plans and goals.	Able to identify and show a **full understanding** of the practical, real-life problem with **comprehensive,** accessible plans and goals	Able to identify and **show understanding** of practical, real-life problem with **good** accessible plan and goals, but may lack some depth or detail	Able to identify and **show some understanding** of practical, real-life problem with **simple** plans and goals	**Barely understand** practical, real-life problem with **fair or incomplete plans and goals**	**Unable to understand** practical, real-life problem with **poor or inaccessible plans and goals**
Formulate strategies and methodologies to achieve project objectives	Able to formulate **comprehensive strategies** and **clear & correct methodologies** to achieve project objectives	Able to formulate **complete strategies** and **correct methodologies** to achieve project objectives	Able to formulate **simple strategies** and **simple methodologies** to achieve project objectives	Barely formulate **fair strategies** and **methodologies** to achieve project objectives	**Unable to formulate** possible strategies and methodologies to achieve project objectives
Select, organize, integrate, and apply the necessary analytical and practical skills to address the problems.	Able to select, organize, integrate, and apply the **most appropriate** analytical and practical skills to **address all the problems effectively and efficiently**	Able to select, organize, integrate, and apply the **necessary** analytical and practical skills to **address all the problems**	Able to select, organize, integrate, and apply **some** analytical and practical skills to **address some of the problems only**	Barely select, organize, integrate, and apply the **limited analytical** and practical skills to address **some of the problems only**	**Unable** to select, organize, integrate, and apply the **limited or inappropriate analytical** and practical skills to address **a few problems**
Apply professional and academic knowledge and skills to solve the defined problem.	Apply the **most appropriate** professional and academic knowledge and skills to solve the defined problem **collaboratively**	Apply **appropriate** professional and academic knowledge and skills to solve the defined problem **collaboratively**	Apply **some** professional and academic knowledge and skills to solve the defined problem **individually**	Apply **limited** professional and academic knowledge and skills to solve the defined problem **partially**	**Unable to apply** professional and academic knowledge and skills to solve the defined problem
Communicate effectively with team members and project supervisor to produce optimal solutions to achieve all the objectives.	Communicate **effectively** with team members and project supervisor to produce **optimal and complete** solutions to **achieve all the objectives**	Communicate **effectively** with team members and project supervisor to produce **good** solutions to **achieve most of the objectives**	Communicate with team members and project supervisor to produce **simple** solutions to **achieve part of the objectives**	**Barely communicate** with team members and project supervisor to produce **limited** solutions to **achieve a few objectives**	**Unable to communicate** with team members and project supervisor to produce **acceptable** solutions to **achieve objectives**
Assess the social impact of the final product to reflect improvement(s) on the existing work/product or introduce new initiatives to improve human life*	Able to assess **fully and positively** the social impact of the final **optimal and complete** product in terms of the improvement on work or to human life	Able to assess **properly** the social impact of the final **good** product in terms of the improvement on work or to human life	Able to assess **simple** the social impact of the final **self-driven or simple** product in terms of the improvement on work or human life	Barely able to assess the social impact of the final **limited incomplete or self-driven** product in terms of the improvement on work or to human life	**Unable to assess** the social impact of the final **incomplete or inappropriate** product in terms of the improvement on work or human life

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Assessment/Course Work	Excellent	Good	Satisfactory	Fair	Poor
Writing Tasks (e.g., Reports)	Demonstrates comprehensive understanding and mastery of the subject. Writing is clear, logical, and free of errors. Analysis is thorough and well-supported by evidence. The report is well-organized and adheres to professional standards.	Shows good understanding with clear and logical writing. Minor errors may exist, but the content is well-supported by evidence. The report is well-structured and meets most professional expectations.	Demonstrates basic understanding with clear writing. Some errors or omissions in analysis. The report is organized but may lack depth or clarity. Meets minimum professional standards.	Shows limited understanding, and writing lacks clarity or contains significant errors. Analysis is superficial or incomplete. The report is poorly organized and falls below professional expectations.	Demonstrates little to no understanding. Writing is unclear or incoherent, and analysis is missing or irrelevant. The report is disorganized and fails to meet professional standards.
Presentations (e.g., Proposal & Final)	Presentation is highly engaging, well-structured, and thoroughly covers all relevant aspects of the project. Delivery is confident, clear, and demonstrates deep understanding. Visual aids and materials are excellent, enhancing communication.	Presentation is well-structured, engaging, and covers key aspects of the project. Delivery is clear, and understanding of the material is evident. Visual aids are well-prepared and effective.	Presentation covers the necessary points but may lack depth or engagement. Delivery is clear but lack confidence or fluency. Visual aids are functional but not enhance communication significantly.	Presentation is disorganized or lacks depth. Delivery is unclear or lacks engagement, with visual aids poorly prepared or ineffective. Demonstrates limited understanding of the material.	Presentation is poorly structured, lacks coherence, and fails to address key points. Delivery is unclear or disengaged, with ineffective or missing visual aids. Little understanding is demonstrated.
Collaboration with Others	Shows excellent teamwork and leadership. Actively contributes to group discussions and takes initiative in problem-solving. Demonstrates respect and understanding of team members' ideas, fostering effective collaboration.	Works well with others and contributes effectively to group discussions and problem-solving. Demonstrates good communication and collaboration skills, but may not consistently take a leadership role.	Contributes to group work but may rely on others for leadership or initiative. Demonstrates adequate communication and collaboration skills but shows limited proactive engagement.	Participates in group work but contributes minimally. Shows poor communication or collaboration skills, and may not engage actively in problem-solving or discussions.	Fails to contribute to group work or causes disruptions. Demonstrates poor communication and collaboration, negatively impacting the team's performance.
Practical Tasks (e.g., Prototype Design & Construction)	Demonstrates exceptional design and technical skills. The prototype is innovative, fully functional, and exceeds project requirements. Testing and troubleshooting are thorough and well-documented.	Demonstrates good design and technical skills. The prototype is functional and meets project requirements. Testing is complete, and most troubleshooting is successful.	The prototype is functional but may show design flaws or lack innovation. Some testing and troubleshooting are documented but may be incomplete or lack detail.	The prototype is incomplete or poorly designed. Testing and troubleshooting are minimal or incomplete, and the project does not fully meet the requirements.	The prototype is non-functional or significantly incomplete. Little to no testing or troubleshooting is documented, and the project fails to meet basic requirements.
Time Management and Organization	Demonstrates excellent time management and organization, meeting all deadlines and proactively planning for each phase of the project. Well-organized work ensures smooth progression.	Shows good time management and organization, meeting most deadlines and having a clear plan. May experience minor delays but manages them effectively.	Meets some deadlines but struggles with organization. The project is somewhat disorganized, leading to delays or rushed work in later phases.	Struggles with time management and misses deadlines. Organization is poor, affecting the quality of work and the team's progress.	Demonstrates very poor time management and organization, frequently missing deadlines and causing significant delays in the project.

Course Assessment Expectations

As we detail the course assessment expectations, it is essential to review the College's and AENG Programme's Educational/Intended Learning Outcomes (ILOs), which shape our students' learning experiences:

College's Education Outcomes

1. Apply general knowledge and skills to think critically and creatively about issues and problems in classroom and community settings;
2. Apply specific knowledge and skills to complete tasks and solve problems with an awareness of context and professional standards;
3. Articulate ideas and thoughts in clear and logical ways to facilitate communication in different contexts.

AENG Programme ILOs:

ILO1. Communicate ideas in a manner appropriate to the circumstance using English and Chinese languages (oral and written) in social and academic contexts;

ILO4. Identify engineering professional and ethical responsibilities and apply these responsibilities in their work;

ILO5. Identify, formulate and solve computing and engineering problems;

ILO6. Design and conduct experiments in the discipline of computing and engineering, as well as to analyse and interpret data;

ILO7. Design a system or process to meet desired needs within realistic constraints, such as economic, environmental, ethical, health and safety, and sustainability;

ILO8. Apply computer science knowledge on analysing and developing computer systems; (CS theme)

ILO9. Apply computer engineering knowledge and technology on electronic and computer system development; (CE theme)

ILO10. Apply electronic engineering knowledge and skills on basic electronic system development. (EE theme)

CCIT4080 as the Final Year Project

CCIT4080 Project on Knowledge Product Development is regarded as the **Final Year Project** of the Associate of Engineering (AENG) programme.

The assessments in this course are therefore intended to demonstrate the **comprehensive knowledge and skills accumulated from previous courses, together with new knowledge and skills acquired independently during the project**.

Students should therefore expect the academic and technical standards of this course to be comparatively high. The project should demonstrate not merely the completion of assigned tasks, but also the ability to integrate knowledge, investigate unfamiliar problems, make appropriate engineering decisions, develop and evaluate solutions, and communicate the resulting work professionally.

The course is intended to prepare students for further education and professional careers, where autonomy, the ability to address complex challenges, independent learning, and the ability to meet high standards are essential.

Students are expected to approach their project diligently, apply their accumulated knowledge appropriately, acquire additional knowledge and skills where necessary, and demonstrate the ability to address real-world engineering problems.

Independent Project Work

Students should **not limit their project work to designated contact hours (i.e., the scheduled project meetings with the supervisor).**

As a general reference for study workload, many courses expect approximately two hours of study outside scheduled contact hours for every contact hour, although the actual amount varies with the course's nature and difficulty. For CCIT4080, the amount of self-directed project work should be **no less than that normally expected in other courses**.

Students and groups are expected to make continuous progress independently **between scheduled project meetings**. Contact hours are primarily intended for:

- reporting and demonstrating progress to the project supervisor;
- receiving feedback and consultation;
- discussing problems and possible solutions;
- conducting meaningful group discussions and collaborative work; and
- facilitating face-to-face group meetings where arranging additional common times may be difficult because of different course timetables.

Scheduled project sessions should therefore not be treated as the only time during which project development takes place.

Use of Resources and Generative Artificial Intelligence

Students are encouraged to make effective use of available resources, including libraries, laboratories, equipment, software tools, technical documentation, **Generative Artificial Intelligence (GenAI)-enabled tools**, and consultation with their project supervisor.

The availability of GenAI and other digital tools can support activities such as information retrieval, idea generation, writing, coding, troubleshooting, and documentation. However, students remain fully responsible for the **accuracy, technical validity, appropriateness, originality, and academic integrity** of the work submitted.

Given the increasing availability and convenience of GenAI-assisted writing and other forms of support, students should understand that expectations for the **quality of documentation, reporting, technical evidence, and presentation are correspondingly higher**.

Well-written or visually polished material alone does not constitute Excellent performance. **Higher performance bands require substantive evidence of engineering work, such as appropriate technical depth, justified design decisions, implementation, testing and validation, critical evaluation, and demonstrated understanding of the submitted work.**

All use of AI tools must comply with the applicable College and course requirements concerning academic integrity, acknowledgement, and declaration of AI-assisted work.

Reference Standard for the Highest Performance Bands

Starting from the 2026/27 academic year, the course will adopt, on a trial basis, an enhanced reference standard when interpreting the upper performance bands.

As CCIT4080 serves as the Final Year Project of the AENG programme, **project work associated with performance in the A range, particularly A+,** would generally be expected to demonstrate an overall quality comparable to a project that could reasonably be presented or submitted to an **external student competition, project showcase, academic event, or equivalent professional setting**.

Participation in an external competition or event is **not required**. The comparison is intended only as a **reference benchmark for the expected level of excellence**.

Depending on the nature of the project, this may be reflected through the overall combination of technical achievement, completeness, originality, appropriate engineering complexity, independent learning, functionality, systematic testing and validation, quality of evidence, **evaluation of results, limitations and engineering trade-offs,** documentation, and professional presentation.

Research-oriented projects or projects that are not naturally suited to external competitions may demonstrate an equivalent level through the quality of their methodology, technical investigation, experimentation, analysis, validation, and academic or professional presentation.

A functioning prototype or completion of the basic stated requirements should therefore **not, by itself, be interpreted as sufficient evidence of Excellent performance**.

At the same time, **a project's** apparent suitability for an external competition does not automatically result in an A or A+ grade. Performance will still be evaluated against the relevant assessment criteria and the overall evidence submitted.

Group Performance and Individual Performance

Students should note that **members of the same project group are not necessarily awarded the same final course grade**.

A substantial proportion of the project is collaborative and is therefore assessed at group level. However, the course also evaluates individual performance through individual assessment components and evidence of individual contribution accumulated throughout the course.

Relevant evidence may include:

- individual technical contribution;
- contribution to reports, documentation, presentations, prototypes, testing, and other deliverables;
- completion or non-submission of required individual assessments;
- quality and consistency of individual work;
- evidence of tasks completed independently;
- ability to explain, demonstrate, and defend one's own contribution;
- progress demonstrated during project meetings;
- meaningful attendance and participation;
- preparedness for discussions with the project supervisor;
- participation in problem-solving and project decision-making;
- communication and cooperation within the group;
- Peer Evaluations;
- Individual Self-Reflection Reports; and
- other relevant evidence available to the project supervisor.

Attendance alone does not constitute meaningful participation. Students are expected to make productive use of the project sessions, contribute to discussions, report progress, respond to questions, and **where appropriate, provide evidence of their work, such as** designs, calculations, drafts, code, photographs, videos, prototype demonstrations, test results, or other project outputs.

To comply with College and course requirements, **prolonged lateness, early departure, or repeated absence will result in penalties in accordance with the applicable College and course requirements.**

Students with valid reasons for absence should follow the applicable College procedures and provide supporting documentation where required.

Where **clear and sustained differences in individual performance** are supported by the available assessment evidence, students within the same group may receive different overall results. **Such differences in assessed performance may result in final grades differing by approximately one to two sub-grades, where justified by the evidence and the marks obtained across the relevant assessment components.**

Where no meaningful difference in individual contribution or performance can be identified, students within the same group would normally be expected to receive similar results for components based principally on the same group work.

Individual differentiation is, therefore, **evidence-based and is not an automatic adjustment applied to every group**.